

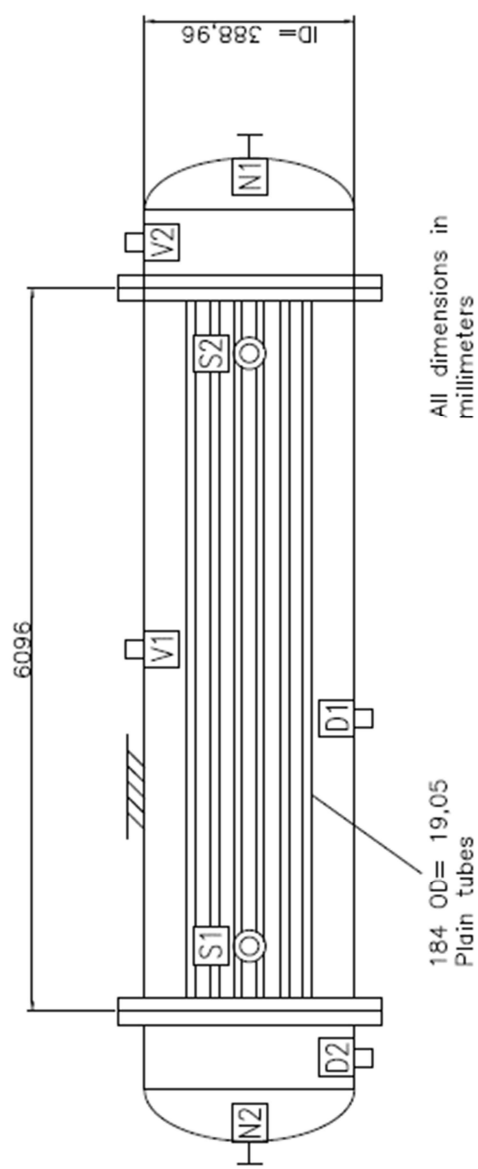
FEED GAS HEATER DATASHEET

IFC	1		ISSUED FOR CONSTRUCTION	LD	MMAR	IC
IFA	0	24-Jun-22	ISSUED FOR APPROVAL	MTOK	MMAR	IC
Status	Rev. n.	Date	Description	Prepared	Checked	Approved
	Rev. Index					
Client			Project Title			
 SOUND ENERGY Mahaj Riad Center, Batiment 6, 3eme etage, Avenue Al Arz, Hay Ryad, 10100 Rabat, Morocco			TENDRARA PROJECT			
Contractor			Provision of a leased train for production of LNG for Sound Energy			
 ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742						
Engineering						
 ITALFLUID GEOENERGY Via Antonio Vivaldi, 4 Montesilvano (PE) ITALY Tel:085-4224374 / fax: 085-4220742						
Technology Vendor			TECHNICAL AMERICA INC			
  TECHNICAL AMERICA A Technical Group Company			301 N Smith Avenue, 92880 Corona (CA)-USA Ph. +1 951 272 9540			
Facility Name		Location		Scale	Sheet of Sheets	
TENDRARA		MOROCCO		A4	1 of 5	
Document Title FEED GAS HEATER DATASHEET				Doc. Code		
				PRO11-ING-FR01		
				Vendor Doc. code		
				21044-PR-DSH-0E201		
Software: Microsoft Excel				File name PRO11-ING-FR01		

Technology vendor		DOC TITLE		FEED GAS HEATER DATASHEET		REV	
		TG DOC NUMBER		21044-PR-DSH-0E201		1	
Customer/ Engineering		ITF DOC NUMBER		PRO11-ING-FR01		PAGE	
		PROJECT		TENDRARA MICRO LNG PLANT		2	

1	Plant/Process Unit Name	LNG Tendirara plant / UNIT 200			Location	Tendirara		
2	Equipment Service	Feed gas heater			Quantity of Units	1		
3	Equipment TAG	E-201			P&ID	PRO13-ING-PR01		
4	Applicable to	☐ Proposal ☒ Construction			As Built	☐ As Built		
5	Purchaser/Agent	TG			Manufacturer/Supplier	CALORIFER		
6	Tema Type	BEM			Serial Number			
7	Information to be completed by	☐ Purchaser ☐ Manufacturer ☐ Manufacturer/Purchaser as Applicable						
GENERAL								
10	Size	16" SCH60	mm	Type	BEM	Connected in	1 parallel	1 series
11	Surf/unit (eff.)	64,506	m²	Shells/unit	1	Surf/shell (eff.)	64,506	m²
PERFORMANCE OF ONE UNIT								
13	Fluid allocation	Shell Side			Tube Side			
14	Fluid Name	Hot oil			Inlet gas			
15	Fluid quantity, Total	kg/h			45566 (Note 3)		18100	
16	Vapor (In/Out)	kg/h					17626 / 17800	
17	Liquid	kg/h			45566		474,37 / 300,47	
18	Noncondensable	kg/h						
19	Temperature (In/Out)	°C			90 / 66,4		11,8 (Note 4) / 60	
20	Dew / Bubble point	°C						
21	Density (Vap / Liq)	kg/m³			/ 838,3		42,609 /	
22	Viscosity	cp			3,46		0,0125	
23	Molecular wt, Vap							
24	Molecular wt, NC							
25	Specific heat	kcal/(kg-C)			0,5265		0,5116	
26	Thermal conductivity	kcal/(h-m-C)			0,1021		0,0285	
27	Latent heat	kcal/kg						
28	Pressure	barg			6,5		45,2	
29	Velocity (Mean/Max)	m/s			0,31 /		4,07 /	
30	Pressure drop, allow./calc.	bar			1 / 0,07		1 / 0,04	
31	Fouling resistance (min)	m²-K/W			0,000216		0,000409	
32	Heat exchanged	kW			558 (Note 7)		MTD corrected 41,7 °C	
33	Transfer rate, Service	207,48			Dirty 260,41		Clean 312,30 W/(m²-K)	
CONSTRUCTION OF ONE SHELL								
35		Shell Side			Tube Side			
36	Design/vac/test pressure:	70 / - / 91			90 / - / 117 barg			
37	Design temperature	-28° / 120°C			-28° / 100°C °C			
38	MDMT				°C			
39	Number passes per shell	1			1			
40	Corrosion allowance	1,5			0 mm			
43	Tube No.	184	OD	19,05 mm	Tks-Avg	2,108 mm	Length	6096 mm Pitch 24 mm
44	Tube type	Plain	Fin n°	#/m	-		Tube pattern	30-Triangular
45	Shell	Carbon Steel	ID	373,08 mm	OD	16" SCH60 mm	Thickness	16,66 mm
46	Channel or Bonnet	Channel cover						
47	Tubesheet-Stationary	Tubesheet-Floating						
48	Tubes in Baffle window	Impingement protection No						
49	Baffle Qty	10	Type	Single Segmental	Cut(%d)	37 V	Spacing: c/c	500 mm
50	Baffle-long	Seal type					Inlet	693,23 mm
51	Supports-tube	Qty			Type			
52	Bypass seal	Tube-tubesheet joint			Expanded (no groove)			
53	Expansion joint	Type						
54	RhoV2-Inlet nozzle	2146,3			Bundle entrance 54,49		Bundle exit 53,53 kg/(ms²)	
55	Gaskets - Shell side	Tube Side						
56	Floating head							
57	Code requirements	ASME Code Sec VIII Div 1			TEMA class		R	
58	Weight/Shell	2648,3			Filled with water 3259,0		Bundle 1051,3 kg	
59	Stamp Required	No			Type			
60	Heat treatment	No			Radiography required		RT3	
62	Overdesign	21% (AREA) For 5°C gas inlet						

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1	NOTES
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3	1 - Test pressure shall be witnessed by authorized Moroccan inspector to allow importation in Morocco.
4	2 - Deleted.
5	3 - Hot oil system is oversized so heat flow will not match with main process balance.
6	4 - The heat exchanger is designed and verified for inlet temperature between 5 and 35°C.
7	This exchanger is designed for winter case and start up.
8	5 - Refer to insulation procedure MEC07-ING-FR02 (21044-QC-INS-00002) and philosophy SAF03-ING-FR02 (21044-QC-INS-00001).
9	6 - Refer to painting procedure MEC07-ING-FR03 (21044-QC-PAI-00001).
10	7 - For minimum inlet temperature 5°C the required duty is 629 kW. Hot oil flow designed for 650 kW.
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